Network Science

Lecture 05: Social Context and Link Formation

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Speaker notes

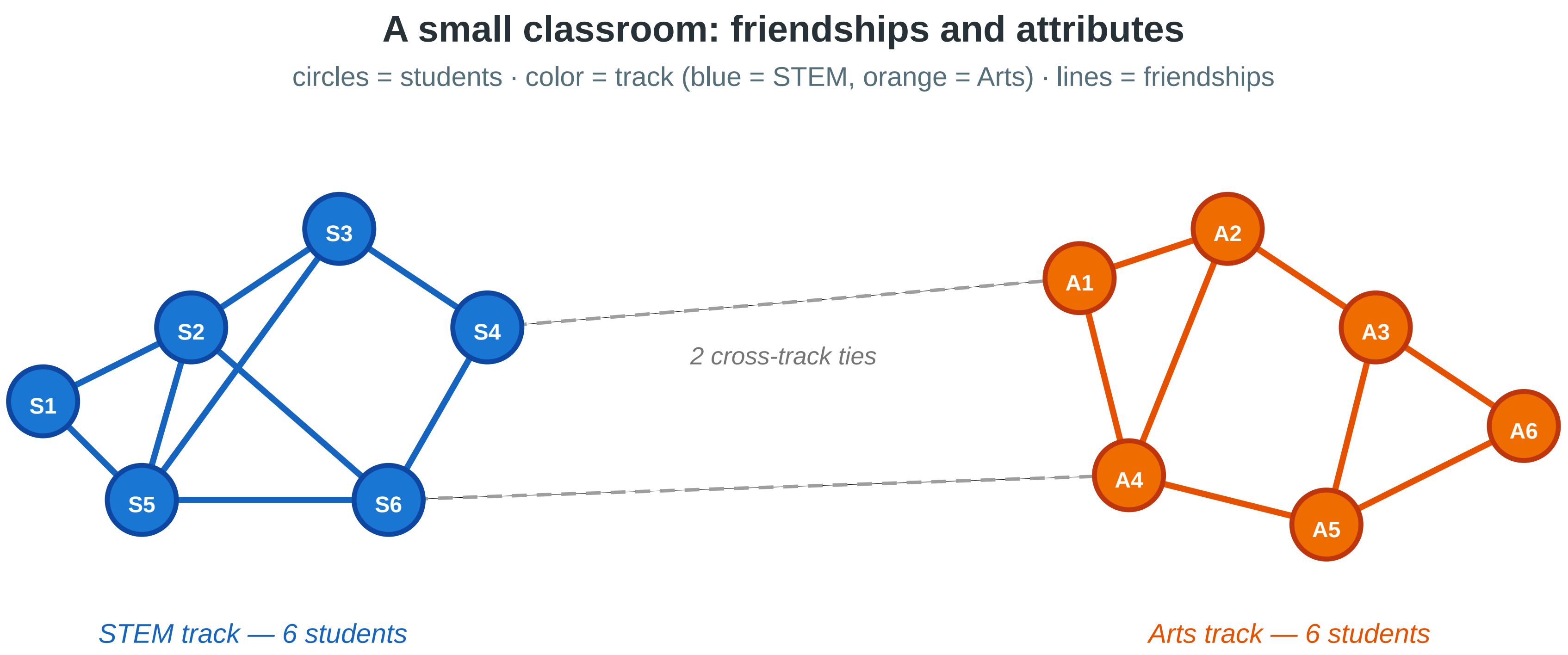
Lectures 03 and 04 treated the graph as a structure of anonymous nodes and unlabeled edges — the only question was "is the edge there?". This lecture adds context: nodes get *attributes* (age, track, beliefs), edges get *timestamps*, and the graph gets *foci* (shared social settings). The central tension shifts from the **computational** gap of L03–L04 (what we want to compute is NP-hard) to a **causal** gap: the mechanism behind a network pattern — selection vs. socialization — cannot be identified from a single snapshot. The fix is not a better algorithm but richer data: longitudinal observations and explicit social contexts.

From Structure to Context

In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario



Questions we'd like to answer from this picture

① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
③ Could shared classes (foci) explain the clustering without any personal preference? ④ Would mild preferences suffice to create this pattern?

Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

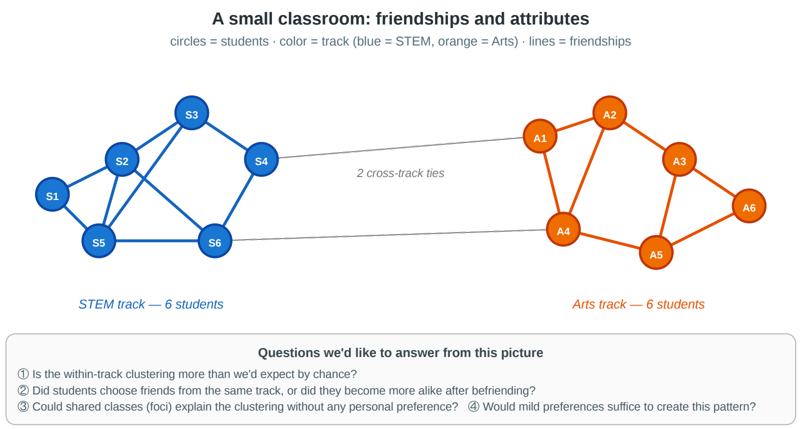
Speaker notes

This figure is the anchor for the whole lecture. Use it to provoke the four questions on the next slide before any definition arrives. By the end of the session students should be able to articulate four different explanations for the visible pattern and say which data would distinguish them.

Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?

Every concept in today's lecture answers one of these.



Speaker notes

Each question maps to a module: (1) homophily and its formal baseline, (2) selection vs. socialization and the causal identification problem, (3) affiliation networks and focal closure, (4) online link formation and (5) Schelling segregation as a model for how mild local preferences generate sharp global patterns. Keep referring back to the classroom figure throughout.

A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)

Speaker notes

This slide is the L05 analogue of L03's "theory vs. measurement" and L04's roadmap table. The new axis is causal identification rather than computational tractability. The rest of the lecture fills in the four rows.

Takeaway

> L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

Speaker notes

This frames the shift cleanly: from computation to causation. Refer back to this framing when each module introduces its specific gap.

Homophily

> **Guiding Question:** Do people form ties because they are similar — and how would we know?

Speaker notes

The key point is that homophily is a *statistical* pattern requiring a counterfactual baseline. "Many within-group ties" is not evidence of homophily unless we know how many ties we would expect under random mixing. Students should leave this module knowing that declaring homophily requires a baseline, not just a visual impression.

Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x. Formally, for a categorical attribute with class shares $p_1, \ldots, p_k$:

- **Random-mixing baseline:** the probability that a random tie connects same-class nodes is

$$H_{\text{base}} = \sum_{i=1}^{k} p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs}.
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

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$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

Speaker notes

The homophily index r is a variant of Newman's assortativity coefficient for categorical attributes (Newman, 2003). It is the standard way to quantify homophily in empirical network studies. The normalization by $(1 - H_{\text{base}})$ matters: in a population that is 90% one type, even random mixing produces 82% within-group ties; reporting 85% same-group without the baseline would overstate the evidence for homophily.

Visual Intuition

Homophily as similarity-biased tie formation

Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Within-group edges (colored) are far more frequent than cross-group edges (gray). But frequency alone is insufficient — we need the random-mixing baseline to decide whether this exceeds chance.

Back to the Classroom

In the scenario, both tracks have 6 students ($p_{\text{STEM}} = p_{\text{Arts}} = 0.5$).

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- The figure has 17 within-track edges out of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

Speaker notes

Walk through the arithmetic on the slide. The punchline — "strong homophily, but the mechanism is unknown" — is the bridge to Module 2. The same $r = 0.78$ would arise under pure selection, pure socialization, or pure focal closure. Distinguishing them requires more than a cross-sectional snapshot.

Back to the Classroom

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The homophily index $r = 0.78$ is strong — far above random mixing. But it does not tell us *why*. Is it selection, socialization, or shared context?

Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. Compute the random-mixing baseline H_{base}.
2. Compute the homophily index r.
3. Is r strong evidence of homophily? What could make it misleading?

Speaker notes

Give them 4 minutes. (1) $H_{\text{base}} = 0.6^2 + 0.4^2 = 0.52$. (2) $r = (0.78 - 0.52)/(1 - 0.52) = 0.26/0.48 \approx 0.54$. (3) $r = 0.54$ is moderate-to-strong, but it can be misleading if the tracks share different physical spaces (focal closure), if students were assigned to tracks based on pre-existing friend groups (reverse causation), or if family background causes both track choice and friendship patterns (confounding).

Quick Check — Solution

$$H_{\text{base}} = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

$$r = \frac{0.78 - 0.52}{1 - 0.52} = \frac{0.26}{0.48} \approx 0.54$$

$r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

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- the two tracks occupy different buildings (shared context, not personal preference)
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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference)
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation)
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding)

Takeaway

> Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

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Selection and Socialization

> **Guiding Question:** How can we distinguish whether similarity caused the tie or the tie caused the similarity?

Speaker notes

This is arguably the most important conceptual module in the lecture. The three mechanisms — selection, socialization, and contextual correlation — produce the same cross-sectional pattern. Distinguishing them requires temporal data and careful causal reasoning. Without longitudinal observations, the $r = 0.78$ we just computed is compatible with any mixture of the three.

Selection and Socialization

Guiding Question: How can we distinguish whether similarity caused the tie or the tie caused the similarity?

$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

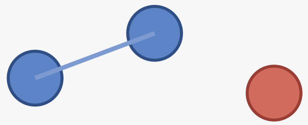
Speaker notes

All three mechanisms are consistent with a high homophily index. A cross-sectional snapshot — friendships and attributes measured at one point in time — cannot separate them. This is the causal wall that L05 is built around.

Causal Diagrams

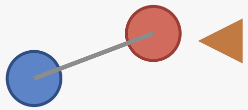
Why observed similarity can arise in different ways

Selection



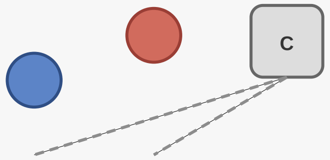
similar actors choose each other

Socialization



connected actors become more alike

Context



a third factor affects both nodes

Speaker notes

Walk through the three causal diagrams: selection (similarity → tie), socialization (tie → similarity), context (third factor → both). Without temporal data, all three explain the same cross-sectional correlation. The point is not that one is always right; in real networks, all three usually operate simultaneously. The analyst's job is to measure their relative contributions.

Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, [New England Journal of Medicine 357, 370–379].

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

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Speaker notes

The Framingham study is the canonical longitudinal analysis of selection vs. socialization. Its strength is the multi-decade time series: because both the network and the attribute (BMI) were measured repeatedly, the authors could observe the temporal order — who became obese *before* whom — and test whether the pattern fit selection, socialization, or confounding. The 57% figure is a relative risk increase among nominated friends compared to non-friends, controlling for geographic proximity.

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Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

What the Study Could Not See

1. Confounders. Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.

2. Homophily in tie formation. The study adjusted for *existing* attribute similarity but could not fully exclude the possibility that people selectively *formed* or *maintained* ties with others on similar weight trajectories — a dynamic form of selection invisible to static controls.

3. Missing network. The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

Speaker notes

This slide is the L05 analogue of L03's "what Kossinets-Watts could not see" and L04's "what Zachary could not see". The pattern is deliberate: each lecture celebrates an empirical result and then identifies its blind spots. The Framingham critique is not a rejection of the study — it is a reminder that even 32-year longitudinal data with repeated measures cannot eliminate all causal alternatives. The lesson is humility, not nihilism.

Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

Speaker notes

Give them 4 minutes. (1) Longitudinal data: friendships and smoking status at multiple time points. (2) Selection: new ties form between people who already both smoke. Socialization: non-smokers start smoking after befriending smokers. (3) Shared environment: dorm floor placement may independently cause both friendship formation and smoking uptake (e.g. shared common rooms where smoking happens).

Quick Check — Solution

Longitudinal observations — friendships and smoking status at multiple time points, with newly formed ties and newly adopted behavior visible.

- : New friendships disproportionately form between people who already share smoking status.
- : People who befriend smokers become more likely to start smoking in the next observation period.

3. Shared — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

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- *Selection*: New friendships disproportionately form between people who already share smoking status.
- *Socialization*: People who befriend smokers become more likely to start smoking in the next observation period.

3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

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Takeaway

> Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

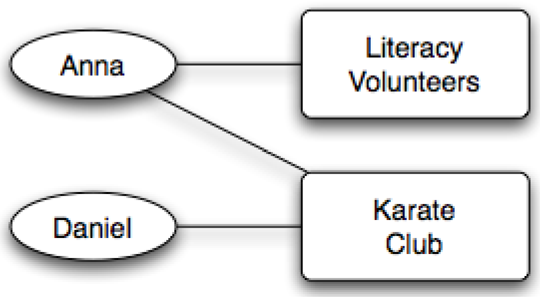
Affiliation Networks

> **Guiding Question:** How do we represent the social settings — classes, clubs, workplaces — in which ties form?

Speaker notes

Affiliation networks make the "shared context" mechanism from Module 2 explicit and formal. Instead of treating context as an unobserved confounder, we model it directly: people are connected to foci, and ties between people arise through their shared foci. This is the modeling answer to Question 3 from the opening slide.

Bipartite Affiliation Model



Source: Easley & Kleinberg 2010

An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

- P: actor nodes (persons)
- F: focus nodes (groups, locations, classes, organizations)
- E: edges only between P and F — "person p participates in focus f"

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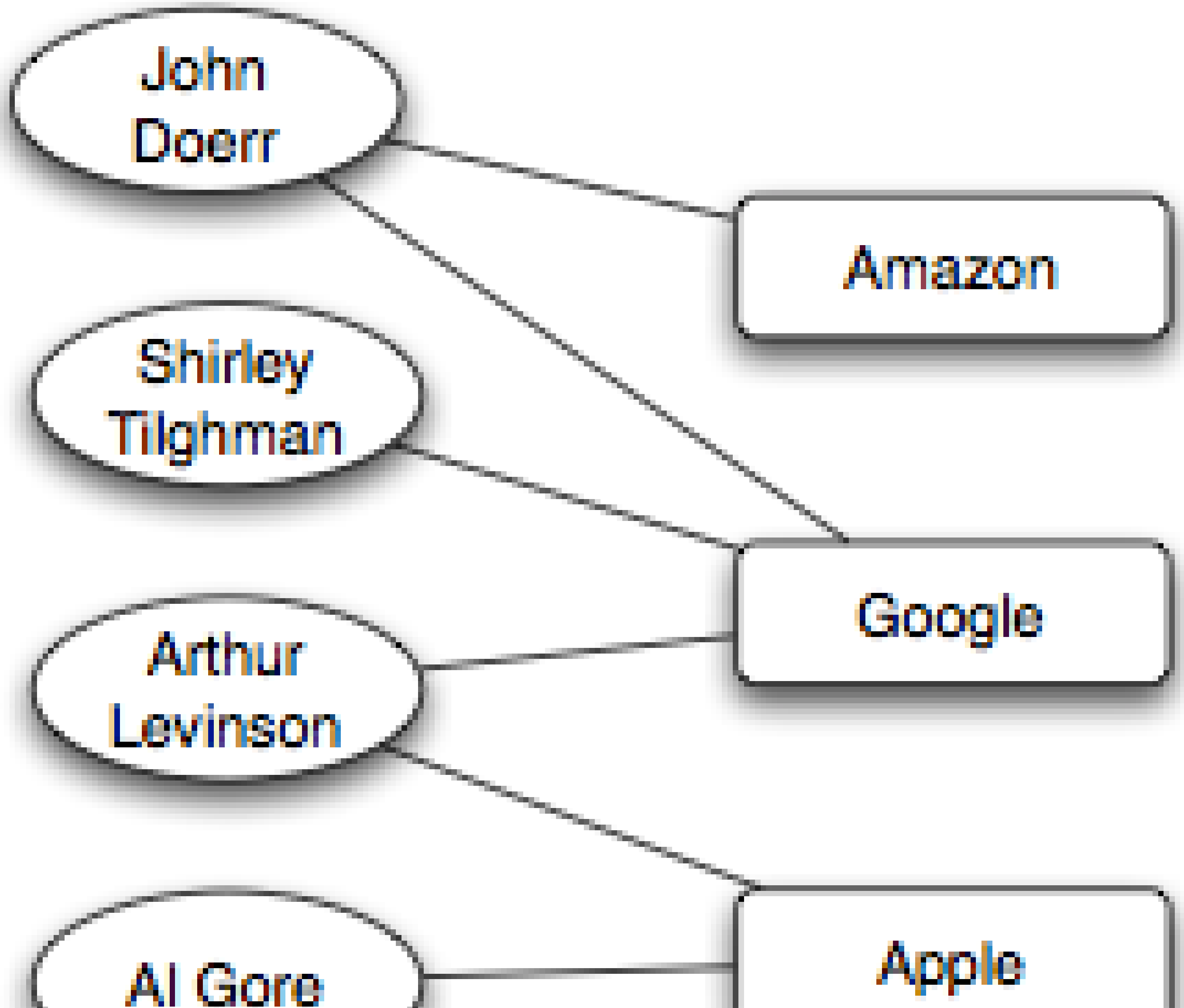
Two persons sharing a focus creates a *potential* for a person–person tie.

Speaker notes

This connects back to Lecture 02's bipartite graphs. The bipartite structure is a modeling decision: we choose to represent contexts as a separate node type rather than as edge attributes. The payoff is that closure processes become explicit and testable. The classroom scenario can be modeled this way: the two tracks are foci, and within-track friendships arise partly through shared-focus exposure.

Example


John Doerr
Shirley Tilghman
Arthur Levinson
Al Gore
Amazon
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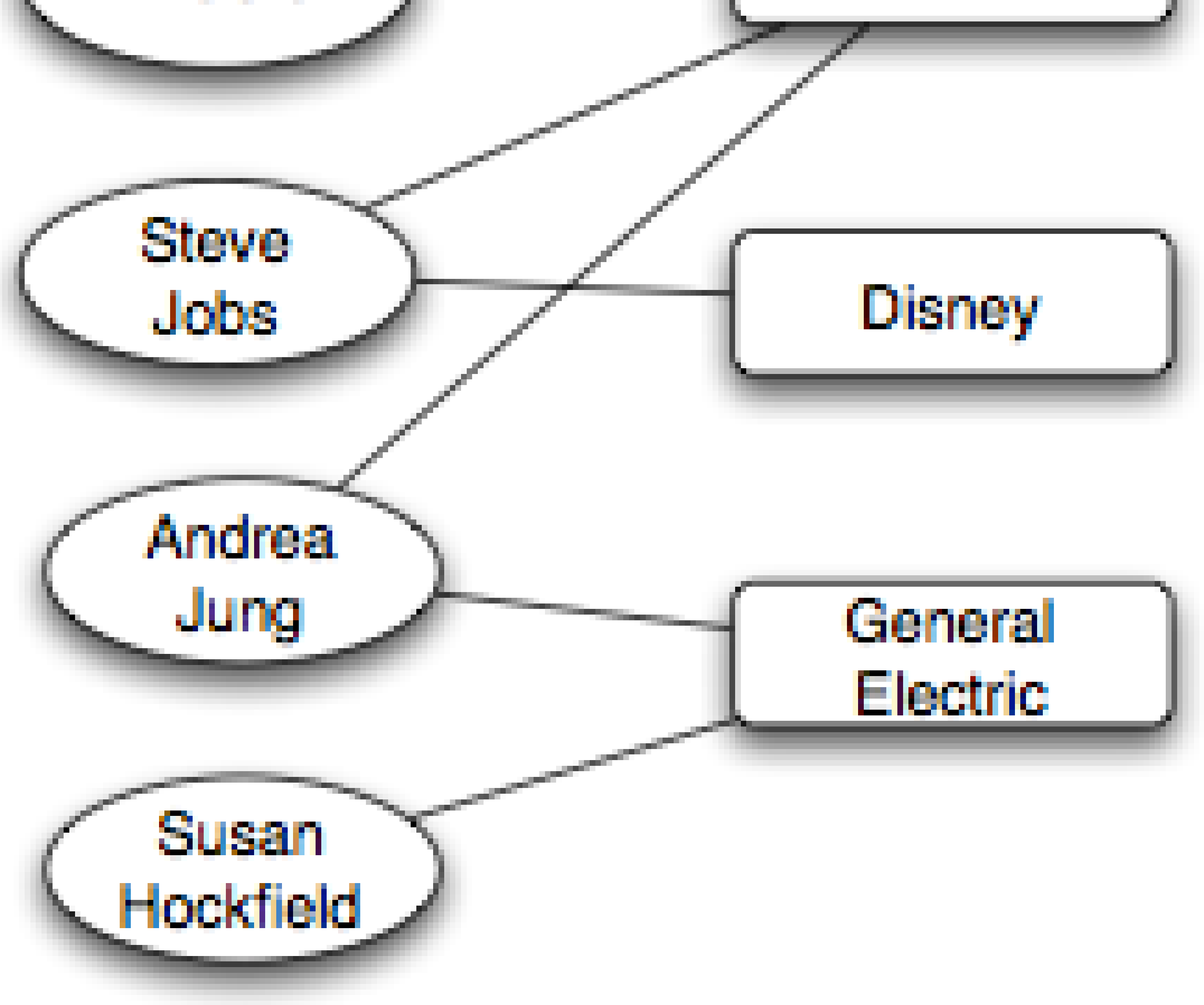
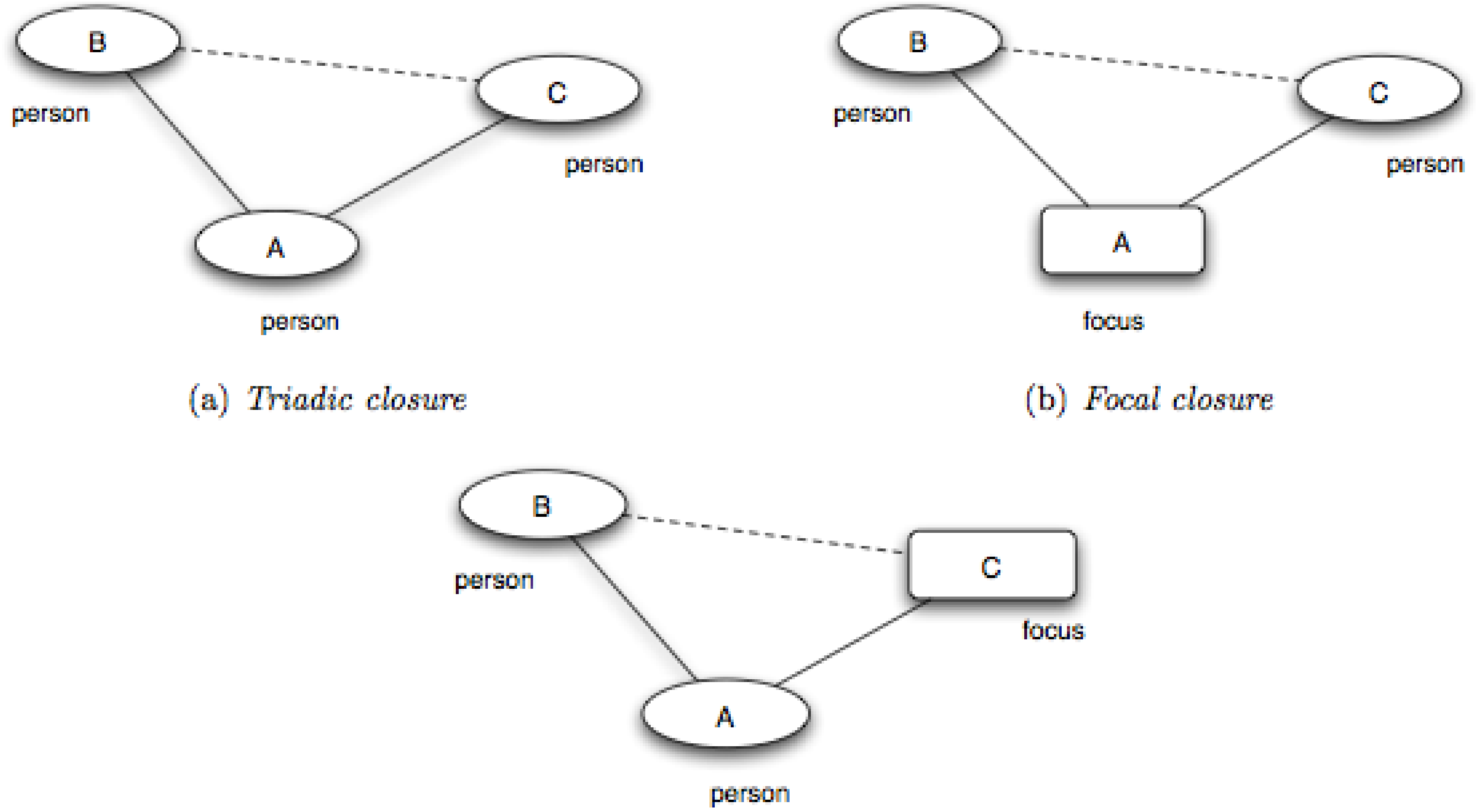


Source: Easley & Kleinberg 2010

Three Closure Processes



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. *(Same mechanism as Lecture 03.)*

Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. *(New — the focus mediates the introduction.)*

Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. *(New — the friend mediates membership.)*

Speaker notes

All three are mechanisms that turn structure into *more* structure. Triadic closure was L03's core concept; focal and membership closure are L05's additions. In the classroom figure, within-track friendships could arise through focal closure (shared labs and lectures bring STEM students together), and if an Arts student (A1) befriends a STEM student (S4) and later joins a STEM elective, that is membership closure. The distinction matters because focal and membership closure produce homophily-like patterns *without* any personal preference for similarity.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Speaker notes

Give them 3 minutes. (1) Focal closure — the elective is the shared focus. (2) Triadic closure — the new friendship forms via a shared friend (the advisor). (3) Membership closure — your friend introduces you to a focus you did not previously belong to.

Quick Check — Solution

1. — the elective is the shared focus that brings them into contact.
2. — the advisor is the shared friend that closes the triangle.
3. — your friend brings you into a focus you did not previously belong to.

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1. **Focal closure** — the elective is the shared focus that brings them into contact.
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Takeaway

Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

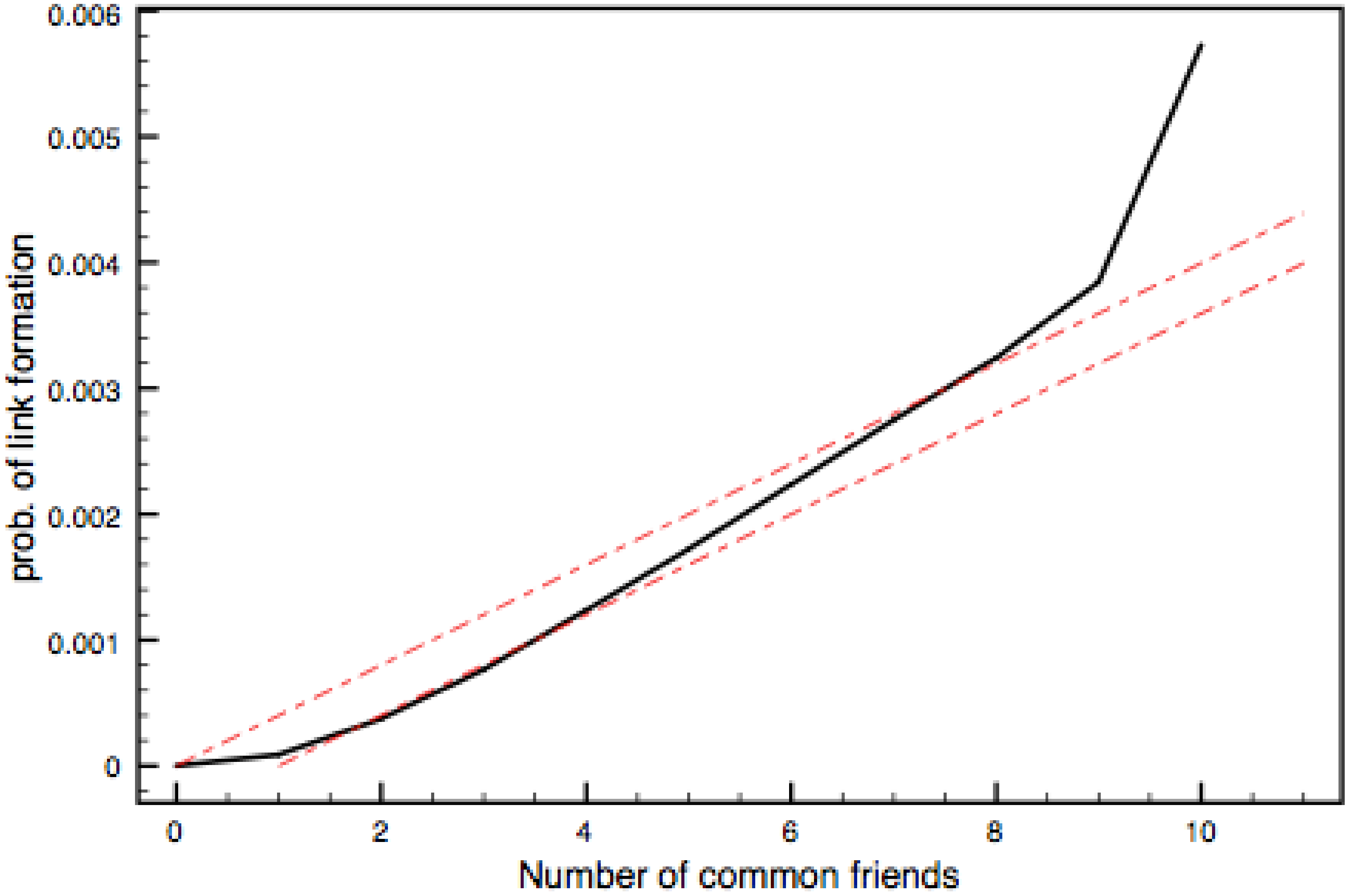
Online Link Formation

> **Guiding Question:** What can large-scale online data teach us about how ties form over time?

Speaker notes

Online platforms make link formation a measurable process at scale. Every new edge has a timestamp, and every preceding state is observable. This is what makes large-scale empirical tests of closure possible — and it is also where we finally get the longitudinal data that Modules 1–2 argued we need.

Triadic Closure at Scale



Source: Easley & Kleinberg 2010

The probability of a new edge rises with the number of shared contacts — but with **diminishing returns**.

Speaker notes

The empirical curve shows that having 1 shared friend already raises the closure probability significantly above baseline, and additional shared friends amplify the effect. But the curve flattens — the 10th shared friend adds far less than the 1st. This rules out a purely additive model.

A Toy Model for the Shape

If each shared contact independently offers an introduction opportunity with probability q, then the probability of closure given k shared contacts is:

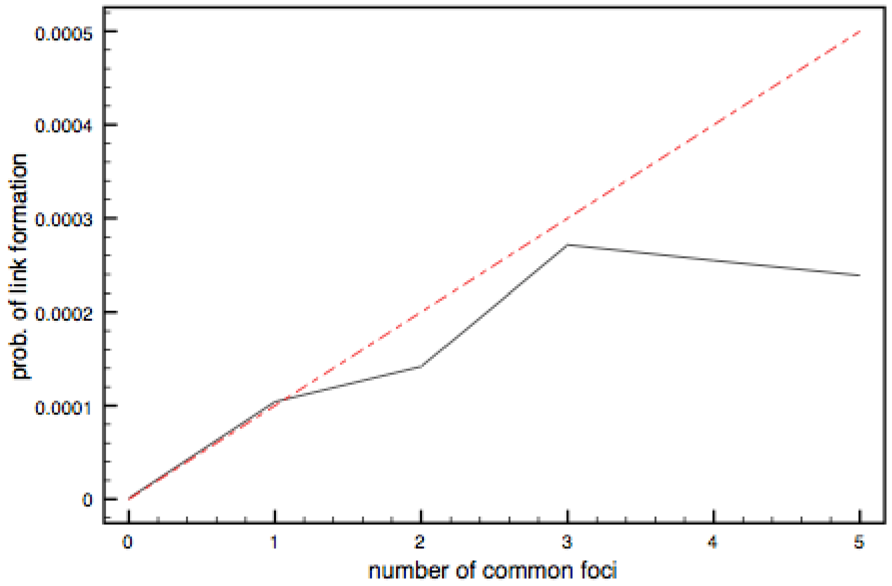
$$P(\text{new tie} \mid k \text{ shared}) = 1 - (1 - q)^k$$

This *saturating* curve rises steeply at first and flattens as k grows — matching the empirical shape qualitatively. But real curves often flatten *faster* than this model predicts, suggesting that shared contacts are not fully independent.

Speaker notes

The $(1-(1-q)^k)$ model is a useful baseline because it is the simplest model that produces saturation. The fact that real data flattens faster than the independent model is itself informative — it suggests redundancy among shared contacts, consistent with the embeddedness idea from L04.

Focal and Membership Closure at Scale



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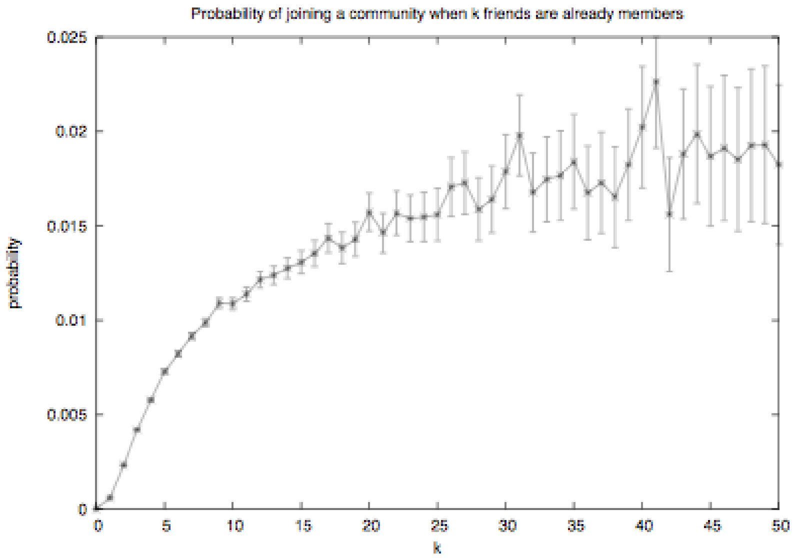

Source: Easley & Kleinberg 2010

Shared foci (left) and friend-mediated memberships (right) both raise the probability of new ties — again with diminishing returns. The shapes are qualitatively similar to triadic closure, reinforcing that all three closure mechanisms share a common structural logic.

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

Speaker notes

Give them 4 minutes. (1) Steep initial rise, then flattening. (2) At $k = 1$: $1 - (1-q)^1 = q$, so $q = 0.20$. But the baseline is 1%, not 0%, so more carefully $P = 0.01 + (1-0.01)(1-(1-q))$ which still gives $q \approx 0.19$. (3) At $k = 2$: the model predicts $1 - (1-0.20)^2 = 0.36$, but we observe only 0.30. The observed curve is *flatter* than the independent model, suggesting shared friends are correlated (redundant), not independent.

Quick Check — Solution

Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%).

At $k = 1, P \approx 0.20 \Rightarrow q \approx 0.20.$

3. At $k = 2$ the independent model predicts $1 - (1 - 0.20)^2 = 0.36$, but we observe 0.30 — a shortfall. The shared friends are (typically embedded in the same group), so they provide less independent opportunity than the model assumes. The diminishing returns are steeper than independence predicts.

Quick Check — Solution

1. Shape: Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%).

At $k = 1, P \approx 0.20 \Rightarrow q \approx 0.20$.

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Quick Check — Solution

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2. Estimating q: At $k = 1, P \approx 0.20 \Rightarrow q \approx 0.20$.

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Takeaway

> Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rathe than independent.

From Local Preferences to Global Segregation

> **Guiding Question:** Can weak local preferences generate strong global segregation?

Speaker notes

This module answers Question 4 from the opening slide: "Would mild preferences suffice to create the sharp pattern we see?" Schelling's threshold model shows that the answer is yes — and the mechanism is iterated local adjustment, not central planning.

Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

Speaker notes

The formal model is simple enough to simulate in a few lines of code, yet its output is dramatically disproportionate to its input. The agent asks for 33% same-type neighbors; the iterated dynamics deliver 90%+ segregation. No agent *wanted* the global outcome; it emerged from the accumulation of individually mild adjustments.

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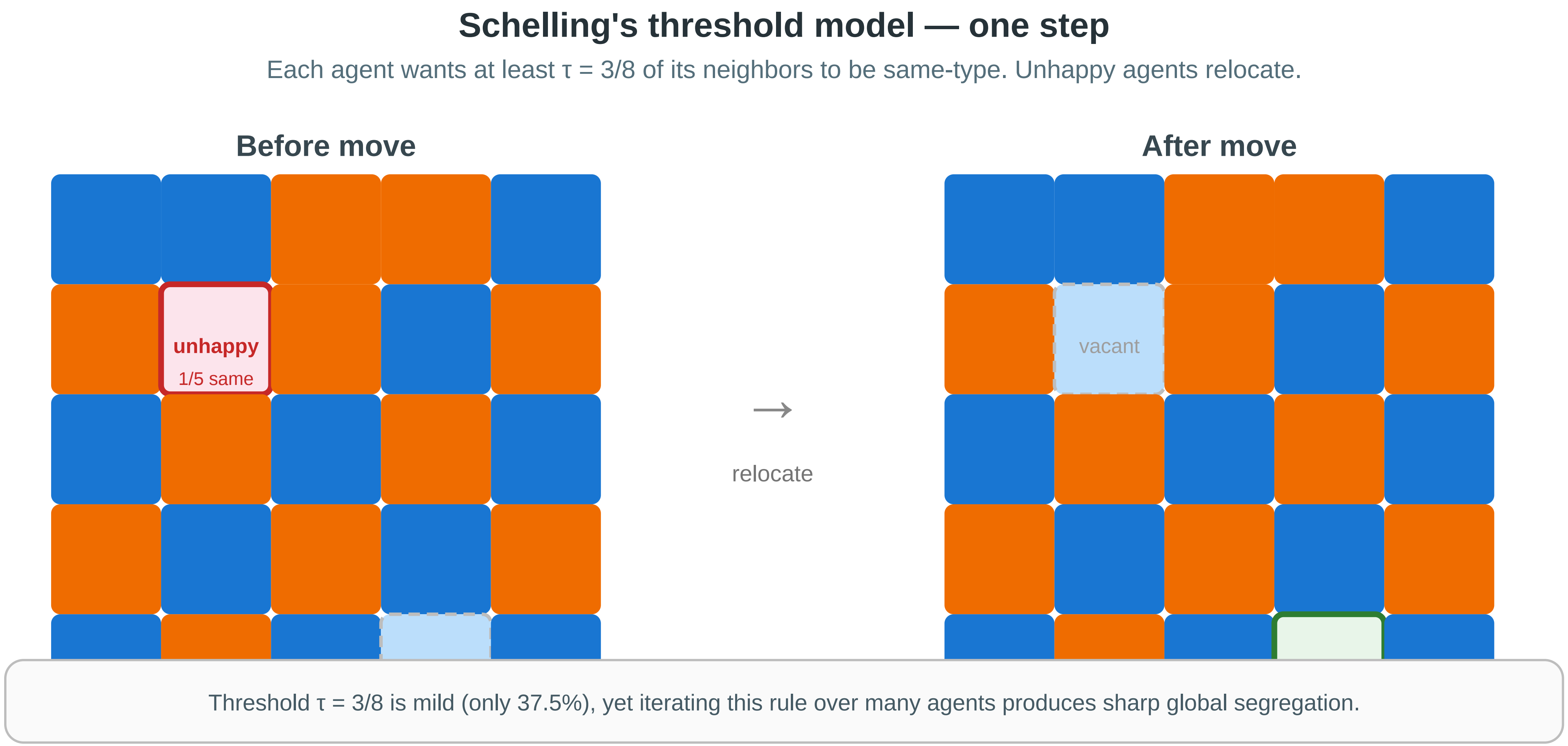
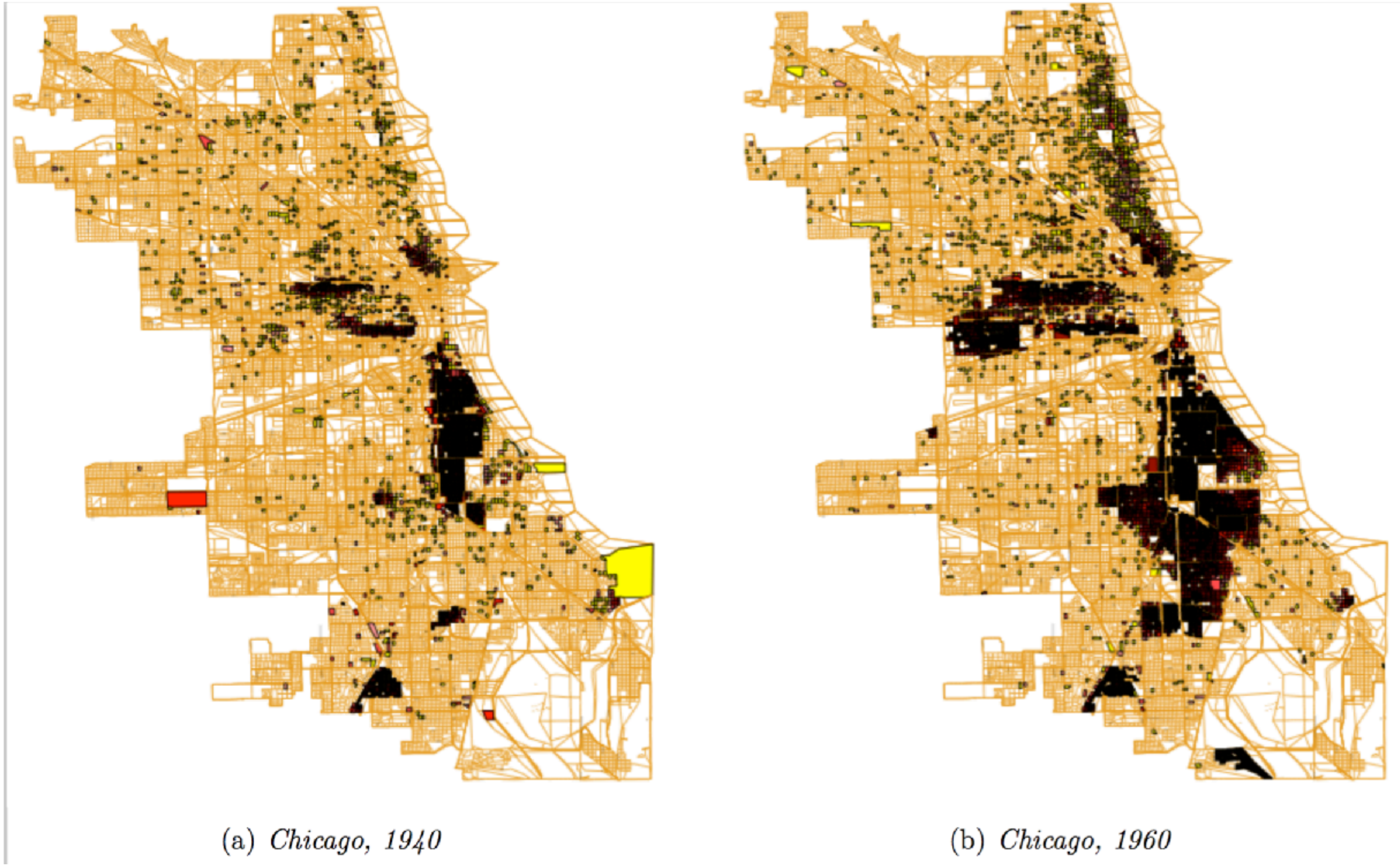
Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

One Step of the Model



The blue agent at (1,1) has 1/5 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

Empirical Motivation



Source: Easley & Kleinberg 2010

Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

Speaker notes

The empirical map raises the puzzle: how can boundaries be *this* sharp if individual preferences are not extreme? Schelling's model answers: iterated local moves amplify mild preferences into sharp global patterns. The same logic applies to online homophily, recommendation systems, and network rewiring.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

Speaker notes

This is the network-science translation of Schelling. The move from a spatial grid to a graph model changes the topology but preserves the core insight: mild local rules plus iterated updates produce sharp global segregation. The recommendation-algorithm connection brings the 1971 model into the 2020s: algorithms now run the Schelling update at machine speed, with consequences for filter bubbles, echo chambers, and political polarization.

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Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

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Speaker notes

Give them 4 minutes. (1) Network neighbors are graph-adjacent nodes. (2) "Moving" becomes rewiring — dropping ties to dissimilar contacts and adding ties to similar ones. (3) The platform acts as a Schelling updater: it identifies "unsatisfied" tie configurations (low similarity) and offers replacements (higher similarity). It amplifies homophily by accelerating the rewiring process.

Quick Check — Solution

A graph-adjacent node — your immediate contacts.

Rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. The platform is an automated Schelling rewirer. It surfaces similar people, which is equivalent to offering each agent a move toward a higher same-type fraction. This homophily — the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Quick Check — Solution

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≡

Takeaway

> Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — The Three Gaps

Across Lectures 03–05, the course has exposed three kinds of gap between what we *want to know* about a network and what the data *actually shows*:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies (clustering, overlap)
L04	Computational	Optimal partition ($\max Q$)	NP-hard	Heuristics (GN, Louvain, Leiden)
L05	Causal	Mechanism (selection vs. socialization)	Not identifiable from snapshots	Longitudinal data, affiliation models

Every concept in the three lectures occupies one cell of this table.

Looking Ahead: Lecture 06

If signs (positive and negative) — not just connection strengths — matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy relations into a graph-theoretic claim with measurable predictions.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Speaker notes

The three-gap table closes the L03–L05 arc. The computational gap and the causal gap are fundamentally different in character: the first is about algorithms, the second is about study design. Both are present in any real network analysis, and the responsible analyst must address both. Lecture 06 opens a new thread — signed graphs and structural balance — which introduces yet another kind of constraint: not computational or causal but *logical* (can the sign pattern be consistent across all triangles?).

Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/